

# DBA Roles and Responsibilities

# Primary Roles of DBA

- To make corporate data available to the end users and decision makers of the organization
- Design:
  - Ensuring the design of the database meets the needs of organization
- Security:
  - Ensure data and access to the data are secure
- Backup:
  - Ensuring that the data can be restored in the event of either human or system failure
- Performance:
  - Ensuring the database and its subsystems are optimized for performance
- Implementation:
  - Proper implementation of the new database system and applications
  - Database Accessibility
  - Capacity issues
  - planning for future expansion requirements

# Roles of DBA...

- In large organization DBA role lies on the following area:
  - DBA's Security Role
  - System Management Role
  - Database Design Role

# Roles of DBA...

- DBA's Security Role:
  - Database and data security before deploy on production
  - Protecting the database
    - Safeguarding the organization's data and this involves preventing unauthorized use of an access of database
    - Need to make security policy
  - Creating and Managing the database users
    - By using proper schemas, roles and privileges
    - Monitor resource usage by individual users

# Roles of DBA...

- DBA's System Management Role:
  - Day to day management of database and its subsystems
  - Troubleshooting
  - Performance Tuning
    - Ongoing task of DBA
    - Database tuning or application tuning or both
    - Test all the code under the new oracle version before you switch over to it
    - Proactive and reactive tuning
  - Monitoring the system
    - Monitoring space
    - Monitor whether batch jobs are finishing as expected
    - Monitoring log files
  - Minimizing the downtime
    - Resume operation in case problem. E.g. disk replacement etc.
  - Estimating requirements
    - OS, DISK, Memory etc.
  - Developing backup and recovery strategy
  - Loading data from old legacy system at reasonable amount of time
  - Observing change management
    - Process of properly migrating new code in an application and DBA need to understand the process that's in place in his organization
    - Patch apply

# Roles of DBA...

- The DBA's Database Design Role:
  - Designing the database
    - Create ER diagram and suggesting the dependencies and candidates for primary key
    - Good design has better performance
  - Installing and upgrading the software
    - Able to give memory and disk requirements to system admin before installing oracle database
    - Patch apply
    - Creating database
      - Test and production
    - Creating database objects
      - Tables, index etc.

# DBA's Job Classification

- planning, design, and development of new database applications, or major changes to existing applications (development DBA, or architect);
- ongoing maintenance of production databases (operations DBA or production DBA);
- management of an organization's data and metadata (data administrator).

# Oracle DBMS

- Oracle Express edition (free of cost) 12C, 18C
- Oracle personal edition
- Oracle standard edition one
- Oracle standard edition
- Oracle Enterprise edition
- License:
  - ASFU (Application Specific Full Use)
  - FU (Full Use)
  - Type
    - Processor perpetual (per license \$47500)
    - Named user plus



- Development env (vendor side)
- UAT env (customer side)
- LIVE (production) env (customer side)